

Multivariable calculus

1. Vectors, matrices

1.1. $\mathbb{R}^n$

Definition 1.1.1 (Two-dimensional real-coordinate space ($\mathbb{R}^2$))

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}$$

Definition 1.1.2 (Three-dimensional real-coordinate space ($\mathbb{R}^3$))

$$\mathbb{R}^3 = \{(x, y, z) : x, y, z \in \mathbb{R}\}$$

x, y, z should be presented such that the coordinate system is right-handed ($\hat{k} = \hat{i} \times \hat{j}$ should have direction according to the right-hand rule).

Definition 1.1.3 (Open set)

An open set U in $\mathbb{R}^n$ satisfies the following equivalent conditions:

- It is the (possibly infinite) union of open balls $B(x, \varepsilon)$
- It is the (possibly infinite) union of open sets
- Every $x \in U$ has some $B(x, \varepsilon) \subseteq U$
- Is its interior: $U^\circ = U$
- Does not contain its boundary: $\partial U \not\subseteq U$
- It is the complement of a closed set

Definition 1.1.4 (Accumulation point)

x is an **accumulation point** of A iff every neighborhood of x (such as $B(x, \varepsilon)$) contains a point of A other than x .

Definition 1.1.5 (Closed set)

A closed set C in $\mathbb{R}^n$ satisfies the following equivalent conditions:

- It is the complement of an open set
- It contains all of its accumulation points
- If a sequence $x_k \rightarrow x$ has all $x_k \in C$, then its limit x is also in C
- It is its closure: $\overline{C} = C$
- It contains its boundary: $\partial C \subseteq C$

1.2. Vector space

Definition 1.2.1 (Vector space)

A vector space over a field F is a set V with the following operations:

- **Vector addition:** $u \in V, v \in V \mapsto (u + v) \in V$, which satisfies the properties of addition
- **Scalar multiplication:** $\alpha \in F, v \in V \mapsto \alpha v \in V$, which satisfies the properties of scalar multiplication

Definition 1.2.2 (Properties of addition)

For all $u, v, w \in V$:

- (A1) Commutativity: $u + v = v + u$
- (A2) Associativity: $(u + v) + w = u + (v + w)$
- (A3) Identity: $\exists 0 \in \mathbb{R}$ s.t. $0 + u = u$
- (A4) Additive inverse: For $u \in F$, $\exists -u \in F$ s.t. $u + (-u) = 0$

Definition 1.2.3 (Properties of scalar multiplication)

For all $\alpha, \beta \in F, v, w \in V$:

- (S1) Associativity: $(\alpha\beta)v = \alpha(\beta v)$
- (S2) Distributivity over scalar addition: $(\alpha + \beta)v = \alpha v + \beta v$
- (S3) Distributivity over vector addition: $\alpha(v + w) = \alpha v + \alpha w$
- (S4) Multiplicative identity: $1v = v$

Theorem 1.2.4

$\mathbb{R}^n$ is a vector space over the field $\mathbb{R}^n$, where we define

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \\ \vdots \end{bmatrix} = (v_1 + w_1, v_2 + w_2, \dots)$$

$$\alpha \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} = \begin{bmatrix} \alpha v_1 \\ \alpha v_2 \\ \vdots \end{bmatrix}$$

Definition 1.2.5 (Standard basis vectors)

The standard basis vectors of a space are the unit vectors that go along the axes of the space. All vectors in that space can be expressed as sums of scalar multiples of the standard basis vectors of that space.

The standard basis vectors of $\mathbb{R}^2$ are $\hat{i}$ and $\hat{j}$, also called e_1 and e_2 .

The standard basis vectors of $\mathbb{R}^3$ are $\hat{i}$, $\hat{j}$, and $\hat{k}$, also called e_1 , e_2 , and e_3 .

Definition 1.2.6 (Vector length)

The **length** or **magnitude** of a vector $\begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix}$ in $\mathbb{R}^n$ is

$$\left\| \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} \right\| = \left\| \begin{bmatrix} v_1 \\ v_2 \\ \vdots \end{bmatrix} \right\| := \sqrt{v_1^2 + v_2^2 + \dots}$$

Definition 1.2.7 (Displacement vector)

The vector from the end of $\vec{A}$ to the end of $\vec{B}$ when their starts are in the same location.

$$\overrightarrow{AB} = \vec{B} - \vec{A}$$

1.3. Dot product

Definition 1.3.1 (Dot product)

Where $\vec{a}, \vec{b} \in \mathbb{R}^n$ and θ is the angle between $\vec{a}$ and $\vec{b}$:

$$\vec{a} \cdot \vec{b} = \sum a_i b_i = |\vec{a}| |\vec{b}| \cos \theta$$

Also called **scalar product** or **inner product**.

Corollary 1.3.1.1

Where $\vec{a}, \vec{b} \in \mathbb{R}^n$ and θ is the angle between $\vec{a}$ and $\vec{b}$:

$$\cos(\theta) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| |\vec{b}|}$$

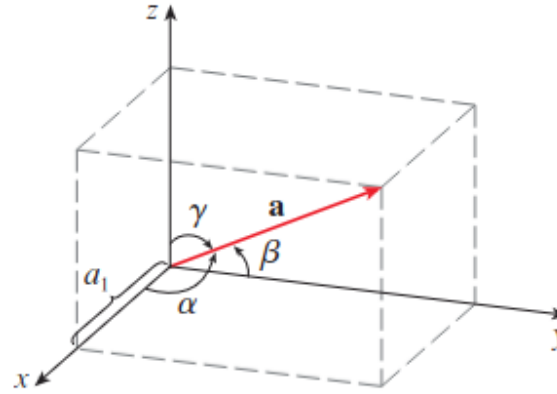
Theorem 1.3.2 (Properties of dot product)

For $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^n$ and $k \in \mathbb{R}$:

- $\vec{a} \cdot \vec{a} = |\vec{a}|^2$
- $\vec{a} \cdot \vec{a} = 0$ iff $\vec{a} = 0$
- Commutativity: $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$
- Distributivity: $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$
- Distributivity: $(k\vec{a}) \cdot \vec{b} = k(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (k\vec{b})$
- $\vec{a} \cdot \vec{b} = 0$ iff $\vec{a} \perp \vec{b}$, $\vec{a} = 0$, or $\vec{b} = 0$.

Definition 1.3.3 (Direction angle)

The **direction angles** of a nonzero vector $\vec{a}$ are the angles $\alpha, \beta, \gamma \in [0, \pi]$ that $\vec{a}$ makes with the positive x, y, z axes respectively.



Theorem 1.3.4

The direction cosines of $\vec{a}$ are the components of the unit vector in the direction of $\vec{a}$:

$$\frac{1}{|\vec{a}|} \vec{a} = \begin{bmatrix} \cos \alpha \\ \cos \beta \\ \cos \gamma \end{bmatrix}.$$

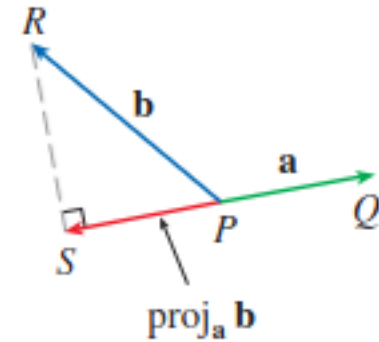
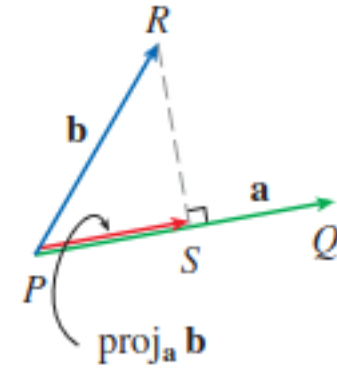
Definition 1.3.5 (Projection)

The **scalar projection of $\vec{b}$ onto $\vec{a}$** or **component of $\vec{b}$ along $\vec{a}$** , is defined as (where $\theta \in [0, \pi]$ is the angle between $\vec{a}$ and $\vec{b}$):

$$\text{comp}_{\vec{a}} \vec{b} := \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} = |\vec{b}| \cos(\theta)$$

The **vector projection of $\vec{b}$ onto $\vec{a}$** , is defined as:

$$\text{proj}_{\vec{a}} \vec{b} := \text{comp}_{\vec{a}} \vec{b} \frac{\vec{a}}{|\vec{a}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|^2} \vec{a}$$



1.4. Cross product

See Stewart 12.4, NMD Lecture 4

Definition 1.4.1 (Cross product)

For $\vec{a}, \vec{b} \in \mathbb{R}^3$, the unique vector $\vec{a} \times \vec{b}$ satisfying

- $|\vec{a} \times \vec{b}|$ is the area of the parallelogram spanned by $\vec{a}$ and $\vec{b}$
- $\vec{a} \times \vec{b} = 0$ iff $\vec{a} \parallel \vec{b}$, $\vec{a} = 0$, or $\vec{b} = 0$.
- $\vec{a} \times \vec{b}$ is orthogonal to $\vec{a}$ and $\vec{b}$.
- $(\vec{a}, \vec{b}, \vec{a} \times \vec{b})$ is right-handed (if the coordinate system is right-handed)

Theorem 1.4.2 (Properties of cross product)

For $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^3$ and $k \in \mathbb{R}$:

- $\vec{a} \times \vec{b} = (-\vec{b}) \times \vec{a}$
- $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$

- $(a + b) \times c = a \times c + b \times c$
- $k(a \times b) = (ka) \times b = a \times (kb)$
- $a \cdot (b \times c) = (a \times b) \cdot c$
- $a \times (b \times c) = (a \cdot c)b - (a \cdot b)c$

Theorem 1.4.3 (Calculation of cross product)

Where $a, b \in \mathbb{R}^3$ and θ is the angle between a and b :

$$\begin{aligned} a \times b &= \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= \hat{i} \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \hat{j} \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \hat{k} \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \\ |a \times b| &= |a| |b| \sin \theta \end{aligned}$$

1.4.1. Parallelograms and parallelepipeds

Theorem 1.4.4 (Parallelogram spanned by vectors in $\mathbb{R}^2$)

The area of the parallelogram spanned by the vectors $\vec{v}, \vec{w} \in \mathbb{R}^2$ is given by

$$\begin{vmatrix} v_1 & v_2 \\ w_1 & w_2 \end{vmatrix} = v_1 w_2 - v_2 w_1.$$

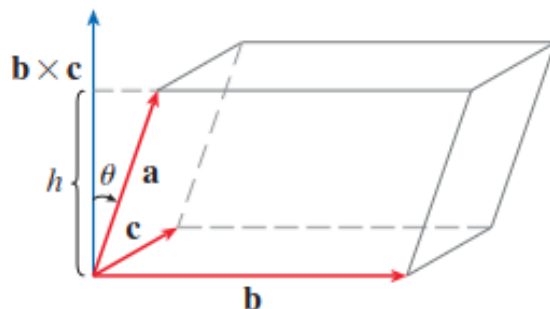
The sign of the result is positive if $\vec{w}$ is counterclockwise from $\vec{v}$ and negative if clockwise.

Definition 1.4.5 (Triple product)

The **triple product** of $\vec{u}, \vec{v}, \vec{w} \in \mathbb{R}^3$ is defined as

$$\begin{aligned} u \cdot (v \times w) &= (u \times v) \cdot w = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix} \\ &= w_1 \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - w_2 \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + w_3 \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix} \end{aligned}$$

Theorem 1.4.6 (Volume of parallelepiped)



Consider the parallelepiped determined by $\vec{a}, \vec{b}, \vec{c} \in \mathbb{R}^3$, shown above. The area of its base is $|b \times c|$. Let θ be the angle between a and $b \times c$; then, its height is $|a| \cos \theta$. Thus, the volume of the parallelepiped is

$$V = Ah = |b \times c| |a| \cos \theta = |a \times (b \times c)|.$$

1.5. Matrices

Definition 1.5.1 (Matrix multiplication)

The matrix multiplication of the $m \times n$ matrix A and the $n \times p$ matrix B is made by dot-producting the rows of the first by the columns of the second:

$$[AB]_{ij} = [A_{i*} \cdot B_{*j}] = \left[\sum_{k=1}^n A_{ik} B_{kj} \right]$$

Definition 1.5.2 (Positive definite)

Let A be a matrix. Then A is positive definite iff any of the following equivalent conditions are true:

- For all $v \in \mathbb{R}^n \setminus \{0\}$, $v^T A v > 0$
- All leading principal minors of A are positive
- All eigenvalues of A are positive
- All pivots of A are positive

Definition 1.5.3 (Negative definite)

Let A be a matrix. Then A is negative definite iff any of the following equivalent conditions are true:

- For all $v \in \mathbb{R}^n \setminus \{0\}$, $v^T A v < 0$
- The k th-order leading principal minor is negative if k is odd and positive if k is even
- All eigenvalues of A are negative
- All pivots of A are negative
- $-A$ is positive definite

Positive semidefinite and **negative semidefinite** are the same except that the determinant/eigenvalue/pivot/ $v^T A v$ could also be 0.

Definition 1.5.4 (Principal minor)

The determinant of a submatrix of a matrix.

Definition 1.5.5 (Leading principal minor)

The k th-order leading principal minor is the determinant of the top left submatrix of a matrix, where the 1st-order leading principal minor is the determinant of the 1x1 matrix at its top left corner, the 2nd-order is the determinant of the 2x2 matrix at its top left corner, etc.

1.6. Useful properties

Theorem 1.6.1 (Cauchy-Schwarz inequality)

For all $\vec{x}, \vec{y} \in \mathbb{R}^n$,

$$|\vec{x} \cdot \vec{y}| \leq |\vec{x}| |\vec{y}|.$$

Theorem 1.6.2 (Triangle inequality)

For all $\vec{x}, \vec{y} \in \mathbb{R}^n$,

$$|\vec{x} + \vec{y}| \leq |\vec{x}| + |\vec{y}|.$$

2. Lines and planes

2.1. Lines

See Stewart [12.5]

The most useful notation for a line is in parametric form:

Definition 2.1.1 (Parametric form of a line)

Where $\vec{r}_0 \in \mathbb{R}^3$ is a point on the line, and $t \in \mathbb{R}^3$:

$$\vec{r}(t) = \vec{r}_0 + \vec{v}t$$

$\vec{v}$ is called the **direction vector**. Then, every value of t gives a point $\vec{r}(t)$ on the line.

Theorem 2.1.2

Two lines are parallel iff their direction vectors are scalar multiples of each other.

Definition 2.1.3 (Skew lines)

Two lines are skew iff they do not intersect but are not parallel, i.e. they lie in different parallel planes.

Procedure 2.1.4 (Finding the intersection of two lines)

Given lines with parametric equations

$$r_1(t) = a_1 + v_1t \quad r_2(t) = a_2 + v_2t$$

solve the system of equations

$$a_1 + v_1t_1 = a_2 + v_2t_2$$

for t_1 or t_2 , then plug it in to the appropriate equation.

(Break up the equation into its x , y , and z components, or whichever is appropriate for your coordinate space.)

2.2. Planes

See Stewart [12.5]

Multivariable calculus 3.1.

Definition 2.2.1 (Plane)

A plane Π is determined uniquely by a point P in the plane and a normal vector n .

A plane is the set of points A in space such that $\overrightarrow{AP}$ is perpendicular to n .

Theorem 2.2.2 (Vector equation for a plane in $\mathbb{R}^3$)

If $\vec{n} \in \mathbb{R}^3$ is the **normal vector** to the plane (vector perpendicular to the plane), and $\vec{P} \in \mathbb{R}^3$ is a point on the plane, then any other point $\vec{r} \in \mathbb{R}^3$ satisfies:

$$\vec{n} \cdot (\vec{r} - \vec{P}) = 0$$

or equivalently:

$$\vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{P}$$

We can also write it as a scalar equation:

$$n_x(x - P_x) + n_y(y - P_y) + n_z(z - P_z) = 0$$

Theorem 2.2.3

Two planes are parallel iff their normal vectors are parallel.

Theorem 2.2.4 (Equation of plane containing three points)

If $A, B, C \in \mathbb{R}^3$ are points on our plane, then we can find the normal vector by performing $\vec{n} = \overrightarrow{AB} \times \overrightarrow{AC} = (B - A) \times (C - A)$ (since $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are vectors on the plane).

Theorem 2.2.5 (Parametric equation for a plane in $\mathbb{R}^3$)

If $a, b \in \mathbb{R}^3$ are nonparallel nonzero vectors on the plane, and $P \in \mathbb{R}^3$ is a point on the plane, then the parametric equation for the plane is:

$$\vec{x}(s, t) = \vec{P} + s\vec{a} + t\vec{b}$$

2.3. Distance

See Stewart [12.5]

Procedure 2.3.1 (Distance between point and line)

Let P be the point, and $A + Lt$ be the line. Then the distance is

$$|\overrightarrow{AP} - \text{proj}_L \overrightarrow{AP}| = \vec{P}$$

Procedure 2.3.2 (Distance between parallel planes)

Let Π_1 and Π_2 be the two planes.

If n is normal to both planes, and $P_1 \in \Pi_1$ and $P_2 \in \Pi_2$, then the answer is

$$|\text{proj}_n \overrightarrow{P_1P_2}|$$

3. Functions and quadric surfaces

3.1. Functions of two variables

See Stewart [12.6]

See Colley [2.1], Trimm [3.1, 3.2], Brummet [MVCWUP:Feb4(35-41)]

Definition 3.1.1 (Function)

All functions $f : X \rightarrow Y$ are defined by:

- A domain set X
- A codomain set Y
- A rule of assignment that associates a unique element $y \in Y$ to each element $x \in X$

Definition 3.1.2 (Graph)

The graph of $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is the set

$$\{(x_1, \dots, x_n, f(x)) : x = (x_1, \dots, x_n)\}$$

Specifically, for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ the graph is the set

$$\{(x, y, z) : (x, y) \in X \text{ and } z = f(x, y)\}$$

Definition 3.1.3 (Level set)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. The **level set at height c of f** is the set in $\mathbb{R}^n$ defined by the equation $f(\vec{a}) = c$, where c is a constant. This is equivalent to the set

$$\{\vec{x} \in \mathbb{R}^n : f(\vec{x}) = c\}$$

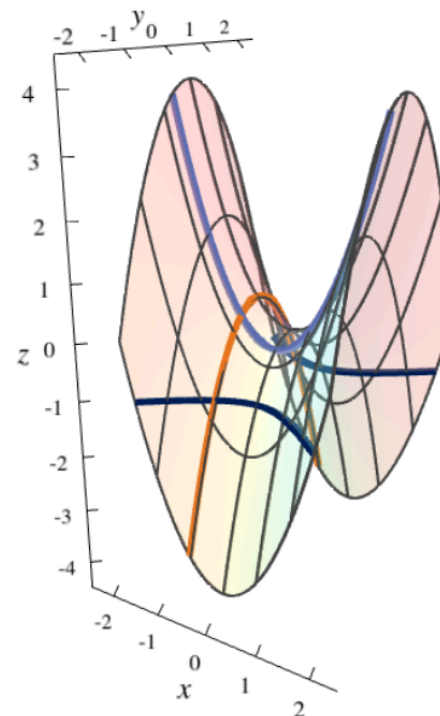
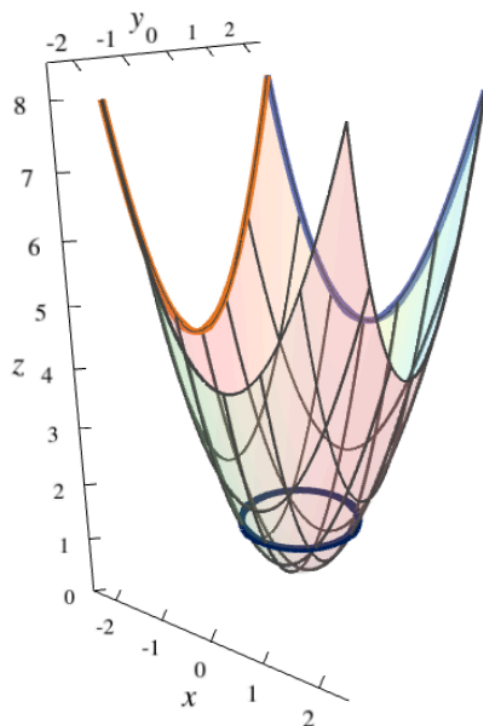
In $\mathbb{R}^2$, this is also called a **level curve**.

Definition 3.1.4 (Contour set)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$. The **contour set at height c of f** is the set in $\mathbb{R}^{n+1}$ defined by the two equations $z = f(\vec{a})$ and $z = c$, where c is a constant. This is equivalent to the set

$$\{\vec{x} \in \mathbb{R}^{n+1} : z = f(\vec{x}) = c\}$$

If $f : X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$, this is also called a **contour curve**. It is equivalent to the level curve, except it is located in $\mathbb{R}^3$ rather than $\mathbb{R}^2$.



3.2. Quadric surfaces

See <https://nathandunfield.github.io/quadrics/index.html>

Definition 3.2.1 (Elliptic paraboloid)

The graph of the equation

$$z = Ax^2 + By^2$$

where A and B have the same sign.

The vertical cross-sections are parabolas, while the horizontal cross-sections are ellipses. Increasing $|A|$ makes it steeper in the x -direction, while increasing $|B|$ makes it steeper in the y -direction.

Definition 3.2.2 (Hyperbolic paraboloid)

The graph of the equation

$$z = Ax^2 + By^2$$

where A and B have opposite signs.

The vertical cross-sections are parabolas, while the horizontal are hyperbolas. Increasing $|A|$ makes it steeper in the x -direction, while increasing $|B|$ makes it steeper in the y -direction.

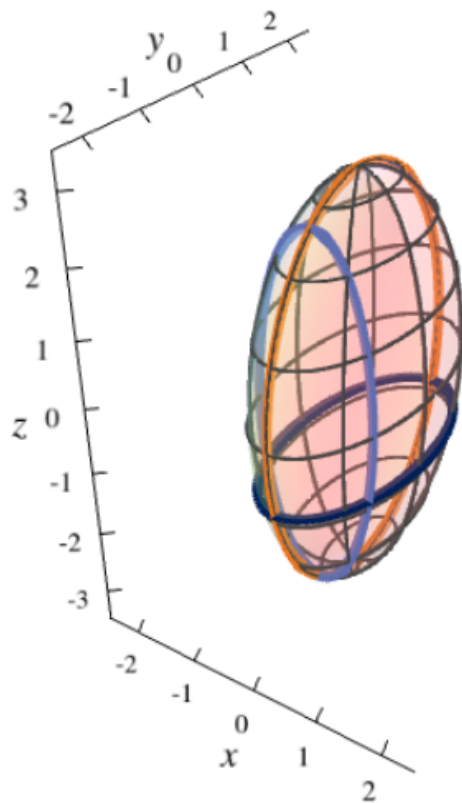
Definition 3.2.3 (Ellipsoid)

The graph of the equation

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} + \frac{z^2}{C^2} = 1.$$

All of its cross-sections are ellipses.

Increasing $|A|$ makes it wider in the x -direction, $|B|$ makes it wider in the y -direction, and $|C|$ makes it wider in the z -direction.



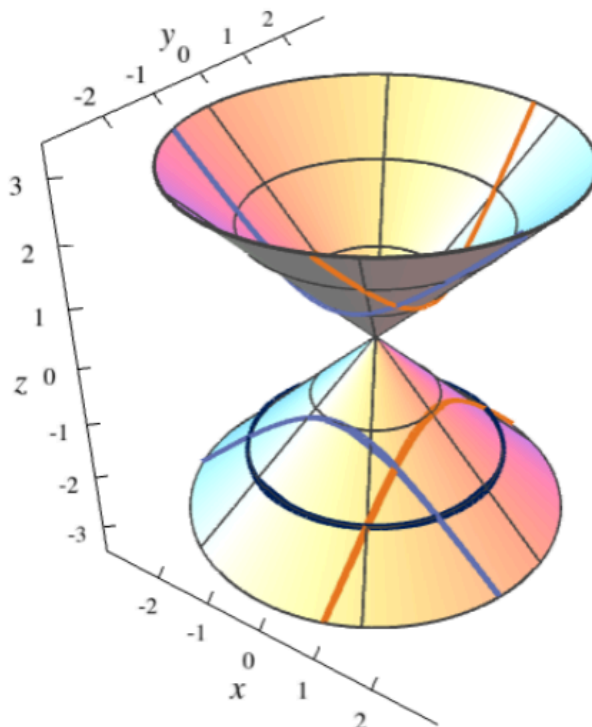
Definition 3.2.4 (Double cone)

The graph of the equation

$$z^2 = Ax^2 + By^2.$$

The vertical cross-sections are hyperbolas, and the horizontal cross-sections are ellipses. Some of its diagonal cross-sections are parabolas. All of its cross-sections are called **conic sections**.

Increasing $|A|$ makes it steeper in the x -direction, while increasing $|B|$ makes it steeper in the y -direction.

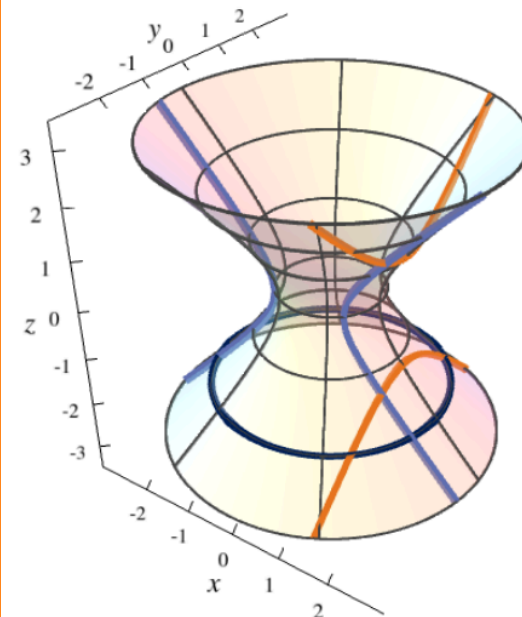


Definition 3.2.5 (Hyperboloid of one sheet)

The graph of the equation

$$\frac{x^2}{A^2} + \frac{y^2}{B^2} - \frac{z^2}{C^2} = 1.$$

The vertical cross-sections are hyperbolas, and the horizontal cross-sections are ellipses. Increasing $|A|$ makes it wider in the x -direction, while increasing $|B|$ makes it wider in the y -direction. Increasing $|C|$ makes it wider in both directions. $|A|$ and $|B|$ affect the “hole” in the middle while $|C|$ does not.



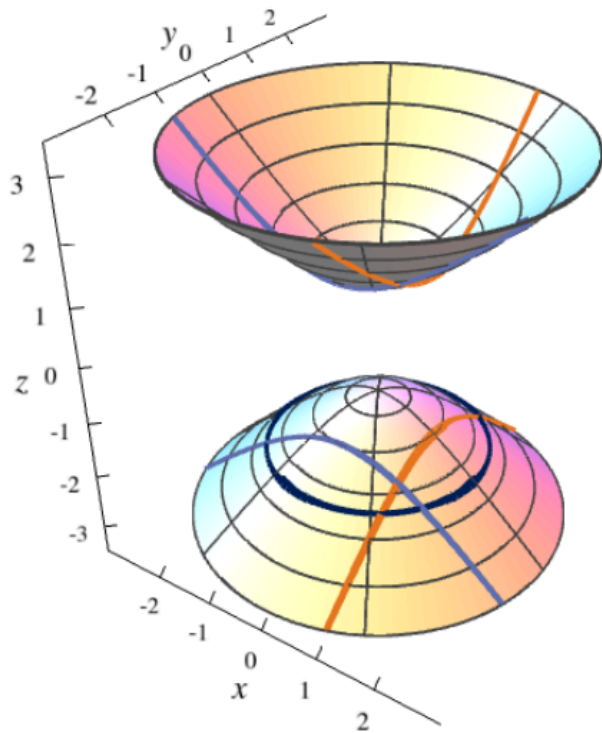
Definition 3.2.6 (Hyperboloid of two sheets)

The graph of the equation

$$-\frac{x^2}{A^2} - \frac{y^2}{B^2} + \frac{z^2}{C^2} = 1.$$

It can be distinguished from the hyperboloid of one sheet by the fact that it is missing some cross-sections; an easy way to check is to check the cross-sections at $x = 0$, $y = 0$, and $z = 0$.

The vertical cross-sections are hyperbolas, and the horizontal cross-sections (where they exist) are ellipses. Increasing $|A|$ makes it wider in the x -direction, while increasing $|B|$ makes it wider in the y -direction. Increasing $|C|$ makes it wider in both directions. $|A|$ and $|B|$ do not affect the “gap” between the top and bottom “sheets”, while $|C|$ does affect the gap.



4. Limits

Definition 4.1 (Limit)

If $f : X \rightarrow \mathbb{R}$, where $X \subseteq \mathbb{R}^n$, then $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = L$ iff for every $\varepsilon > 0$ there exists $\delta > 0$ s.t. $\vec{x} \in X$ and $0 < |\vec{x} - \vec{a}| < \delta$ implies $|f(\vec{x}) - L| < \varepsilon$.

Warning: The limit might be different along different paths to $\vec{a}$! If different paths result in different limits, then there is no limit.

Definition 4.2 (Continuity)

If $f : X \rightarrow \mathbb{R}$, where $X \subseteq \mathbb{R}^n$, then f is **continuous** at $\vec{a}$ iff

$$\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = f(\vec{a}).$$

f is **continuous** on X iff f is continuous at every $\vec{a} \in X$.

Theorem 4.3 (Limit laws)

$$\begin{aligned} \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) + g(\vec{x}) &= \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) + \lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) \\ \lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x})g(\vec{x}) &= \left(\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) \right) \left(\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) \right) \\ \lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x})}{g(\vec{x})} &= \frac{\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x})}{\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x})} \end{aligned}$$

Theorem 4.4 (Squeeze theorem)

If

- $g(\vec{x}) \leq f(\vec{x}) \leq h(\vec{x})$ for all $\vec{x}$ in an open ball containing $\vec{a}$, except possibly at $\vec{a}$ itself, and
- $\lim_{\vec{x} \rightarrow \vec{a}} g(\vec{x}) = \lim_{\vec{x} \rightarrow \vec{a}} h(\vec{x}) = L$,

then $\lim_{\vec{x} \rightarrow \vec{a}} f(\vec{x}) = L$.

5. Differentiation

5.1. Partial derivatives

See Colley [2.3, 2.4], Trimm [3.4, DiffEq-1.0], Brummet [11, 12.5, MVCWUP:Feb12/22(48-54)]

See NMD [8, 9], Stewart []

Definition 5.1.1 (Partial derivative with respect to x)

The partial derivative of $f(x, y)$ with respect to x is

$$\lim_{h \rightarrow 0} \frac{f(a + h, b) - f(a, b)}{h}$$

Let $z = f(x, y)$. Then the partial derivative is denoted by

$$f_x(x, y) = f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = D_x f$$

Definition 5.1.2 (Partial derivative)

The partial derivative of $f(\vec{x})$ with respect to the i th variable is

$$\frac{\partial f(\vec{x})}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f\left(\begin{bmatrix} x_0 \\ \vdots \\ x_i + h \\ \vdots \\ x_n \end{bmatrix}\right) - f(\vec{x})}{h}$$

This is equivalent to letting $F(x_i) = f(\vec{x})$ and finding $F'(x_i)$.

Definition 5.1.3 (Higher-order partial)

The result of taking the partial derivative of a partial derivative, which may be higher-order.

A partial derivative that is not higher-order is called a **first-order partial**. A partial derivative of a first-order partial is a second-order partial, a partial derivative of a second-order partial is a third-order partial, etc.

A higher-order partial which is the result of taking the partial with respect to x_1 , then with respect to x_2 , then with respect to $x_3, \dots, x_n$, is denoted by

$$f_{x_1 \dots x_n} = \frac{\partial}{\partial x_n} \dots \frac{\partial}{\partial x_1} f$$

$x_1 \dots x_n$ do not have to be distinct. If $x_1 \dots x_n$ are not all the same then the higher-order partial is called a **mixed partial derivative**.

Definition 5.1.4 (C^k function)

Where k is a nonnegative integer, a function $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is of order C^k at point $\vec{x} \in X$ iff its k -th order and lower partials exist and are continuous at $\vec{x}$.

It is of order C^∞ at point $\vec{x}$ iff it is of order C^k at $\vec{x}$ for all $k \in \mathbb{N}$.

It is of order C^k iff it is of order C^k at all $x \in \vec{x}$.

Theorem 5.1.5 (Clairaut's theorem)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ whose k -th order and lower partials exist and are continuous on X . Then its k -th order and lower partials may be evaluated in any order, i.e.

$$f_{x_1 \dots x_n} = f_{x_n \dots x_1} = f_{x_1 x_3 x_2 x_7 \dots x_4} = \dots$$

5.2. Differentiability

See Colley [2.3], Trimm [3.5], Brummet [13.5]

See Stewart [14.4], NMD [9]

Definition 5.2.1 (Linear map)

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **linear map** iff $f(\vec{x}) = A\vec{x}$ for some $1 \times n$ constant matrix A .

Definition 5.2.2 (Affine map)

$f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a **affine map** iff $f(\vec{x}) = A\vec{x} + b$ for some $1 \times n$ constant matrix A and constant scalar $b \in \mathbb{R}$.

The **differential** of an affine function is the associated linear function $df(\vec{x}) = A\vec{x}$.

An affine function $f : \mathbb{R} \rightarrow \mathbb{R}$ is of the form $f(x) = ax + b$. Its differential is $df(x) = ax$.

An affine function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ is of the form $f(x) = ax + by + cz + d$. Its differential is $df(x) = ax + by + cz$.

Definition 5.2.3 (Differential)

The **differential** of $f : \mathbb{R} \rightarrow \mathbb{R}$ at a is

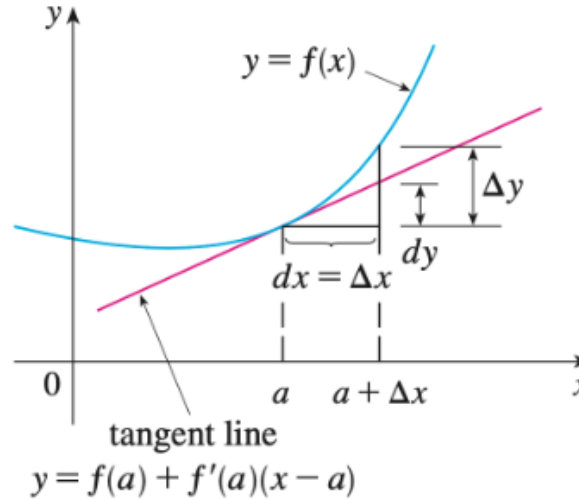
$$df_a(x) := f'(a)x.$$

The differential of $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is

$$\nabla f(\vec{a}) \cdot (\vec{x} - \vec{a}) = \sum \frac{\partial f_i}{\partial x_i}(\Delta x_i) = \sum \frac{\partial f_i}{\partial x_i}(x_i - a_i).$$

Definition 5.2.4

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $y = f(x)$. For small values of Δx , $\Delta y \approx df_a(\Delta x) = f'(a)\Delta x$:



For the sake of convenience, we define $dx := \Delta x$ and

$$dy := df_a(dx) = f'(a)dx = \frac{dy}{dx}dx.$$

The **linear approximation** of f at a is

$$\begin{aligned} f(a + \Delta x) &\approx h(x) := f(a) + f'(a)\Delta x \\ &= f(a) + f'(a)(x - a) \\ &= f(a) + dy. \end{aligned}$$

Definition 5.2.5 (Linear approximation ($\mathbb{R}^n \rightarrow \mathbb{R}$))

The **linear approximation** or **tangent plane** ($\mathbb{R}^3$) or **hyperplane** to the graph of a function f at the point $\vec{a}$ is expressed by

$$L(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a})$$

In $\mathbb{R}^3$, this is equivalent to the plane

$$\begin{aligned} z &= L(x, y) = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b) \\ &= f(\vec{a}) + f_x(\vec{a})\Delta x + f_y(\vec{a})\Delta y. \end{aligned}$$

Definition 5.2.6 (Linear approximation ($\mathbb{R}^n \rightarrow \mathbb{R}^m$))

The **linear approximation** to a vector-valued function f at the point $\vec{a}$ is expressed by

$$L(\vec{x}) = f(\vec{a}) + Jf(\vec{a})(\vec{x} - \vec{a})$$

Definition 5.2.7 (Differentiability)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$, where X is an open subset of $\mathbb{R}^n$, and let $\vec{a} \in X$. f is differentiable at a iff all of its partial derivatives exist and

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{f(\vec{x}) - L(\vec{x})}{|\vec{x} - \vec{a}|} = 0$$

where $L(\vec{x})$ is the linear approximation to f at $\vec{a}$.

If $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, then we can instead say f is differentiable iff

$$\lim_{\vec{x} \rightarrow \vec{a}} \frac{\Delta z - (f_x(\vec{a})\Delta x + f_y(\vec{a})\Delta y)}{|\vec{x} - \vec{a}|} = 0.$$

Theorem 5.2.8

Differentiability implies continuity.

Theorem 5.2.9 (Differentiability shortcut)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a vector-valued function. If all partial derivatives $\frac{\partial f_i}{\partial x_j}$ exist and are continuous in an open ball around $\vec{a}$ in X , then F is differentiable at $\vec{a}$.

5.3. Chain rule

See Colley [2.5], Trimm [3.8, 6.5, DiffEq-1.0], Brummet [12, MVCWUP:Feb24(58-61)]

See NMD [10], Stewart [14.5]

Theorem 5.3.1 (Multivariable chain rule)

Suppose $X \subseteq \mathbb{R}^m$ and $T \subseteq \mathbb{R}^n$ are open and $f : X \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^p$ and $r : T \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ are defined so that $r(T) \subseteq X$. We wish to find the derivative of

$$f \circ r : T \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m \rightarrow \mathbb{R}^p.$$

If r is differentiable at $t_0 \in T$ and f is differentiable at $x_0 = r(t_0)$, then the composite $f \circ r$ is differentiable at t_0 , and we have

$$J(f \circ r)(t_0) = Jf(x_0)Jr(t_0).$$

When $p = 1$ and $n = 1$ (i.e. $f \circ r : T \subseteq \mathbb{R} \rightarrow \mathbb{R}^m \rightarrow \mathbb{R}$), this reduces to

$$(f \circ r)'(t_0) = \nabla f(x_0) \cdot r'(t_0).$$

When $p = 1$ (i.e. $f \circ r : T \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m \rightarrow \mathbb{R}$), define $u := f(x_1, \dots, x_m)$ where $x_j := r_j(t_1, \dots, t_n)$. Then for each $i = 1, 2, \dots, n$,

$$\frac{\partial u}{\partial t_i} = \frac{\partial u}{\partial x_1} \frac{\partial x_1}{\partial t_i} + \frac{\partial u}{\partial x_2} \frac{\partial x_2}{\partial t_i} + \dots + \frac{\partial u}{\partial x_m} \frac{\partial x_m}{\partial t_i}.$$

In $\mathbb{R}^3$, if $n = 1$, $p = 1$, and $m = 3$ (i.e. $f \circ r : T \subseteq \mathbb{R} \rightarrow \mathbb{R}^3 \rightarrow \mathbb{R}$), define $(x, y, z) := (x_1, x_2, x_3)$.

Then,

$$\frac{d(f \circ r)}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}.$$

5.4. Gradient and Jacobian

Definition 5.4.1 (Derivative of vector-valued function)

Let $f : T \subseteq \mathbb{R} \rightarrow \mathbb{R}^m$. Then

$$f'(t) = \begin{bmatrix} f_1'(t) \\ f_2'(t) \\ \vdots \\ f_m'(t) \end{bmatrix}$$

Definition 5.4.2 (Gradient)

$$\nabla f(\vec{x}) = \begin{bmatrix} \frac{\partial f(x)}{\partial x_1} \\ \vdots \\ \frac{\partial f(x)}{\partial x_n} \end{bmatrix}$$

Theorem 5.4.3

$\nabla f(\vec{a})$ is orthogonal to the level set of f at $\vec{a}$.

Definition 5.4.4 (Jacobian)

If $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a vector-valued function, then the Jacobian is

$$Jf \left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \right) = \begin{bmatrix} \nabla f_1 \\ \nabla f_2 \\ \vdots \\ \nabla f_m \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} & \dots & \frac{\partial f}{\partial x_n} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \dots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \dots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

The Jacobian Jf is also called Df or ∇f .

5.5. Directional derivative

See Colley [2.6], Trimm [3.7], Brummet [14]

See NMD [11]

See <https://youtu.be/7JEWlffoJJQ>

Definition 5.5.1 (Directional derivative)

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, where X is an open subset of $\mathbb{R}^n$, and let $\vec{a} \in X$. If $\vec{v}$ is any unit vector in X , then the directional derivative of f at $\vec{a}$ in the direction of $\vec{v}$ is

$$D_{\vec{v}}f(\vec{a}) = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{v}) - f(\vec{a})}{h}$$

We may wish to normalize the directional derivative:

$$D_{\vec{v}}f(\vec{a}) = \lim_{h \rightarrow 0} \frac{f(\vec{a} + h\vec{v}) - f(\vec{a})}{h|\vec{v}|}$$

Theorem 5.5.2

If f is differentiable at $\vec{a}$, then

$$D_{\vec{v}}f(\vec{a}) = \nabla f(\vec{a}) \cdot \vec{v}$$

We may wish to normalize the directional derivative:

$$D_{\vec{v}}f(\vec{a}) = \nabla f(\vec{a}) \cdot \frac{\vec{v}}{|\vec{v}|}$$

Theorem 5.5.3

The gradient points in the direction of steepest ascent, and its magnitude is the derivative in that direction:

$$|\nabla f(\vec{a})| = D_{\widehat{\nabla f(\vec{a})}}f(\vec{a}) = \max\{D_{\vec{v}}f(\vec{a}) : \vec{v} \in \mathbb{R}^n\}$$

where $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$.

Theorem 5.5.4

Let $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$, and let $(\vec{a}, k) \in \mathbb{R}^{n+1}$. Then $\nabla f(\vec{a})$ is orthogonal to the level curve at height k .

Thus, the tangent plane to the surface described by $f(\vec{x}) = k$ at point $\vec{a}$ is the plane with normal vector $\nabla f(\vec{a})$.

Example. The electric potential V 's negative gradient is called $\vec{E}$:

$$\vec{E} = -\nabla V.$$

This is because in electrostatics, opposites attract, so positive charges want to follow the steepest descent to an area with low electric potential. ∇V points to the steepest ascent, so $-\nabla V$ points to the steepest descent.

Theorem 5.5.5

The second directional derivative of f in the direction of $\vec{v}$ is given by

$$v^T(Hf)v.$$

6. Extrema

See Colley [4.1], Brummet [15, MVCWUP:69-80(Mar 4-6)]

See NMD [12, 13], Stewart [14.7]

See <https://youtu.be/7JEWlFfoJJQ>

6.1. Local extrema

Definition 6.1.1 (Critical point of f)

A point $\vec{c}$ in the domain of f where all of the partial derivatives of f at $\vec{c}$ equal 0 or do not exist, or equivalently $\nabla f(\vec{c}) = 0$ or does not exist.

Definition 6.1.2 (Saddle point)

A critical point that is not a max or min.

Theorem 6.1.3 (by Fermat's Theorem)

If $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ has a local maximum or minimum at $\vec{a}$ and the first order partial derivatives exist, then $\vec{a}$ is a critical point.

Definition 6.1.4 (Hessian matrix)

The Hessian matrix Hf of a function $f : X \subseteq \mathbb{R}^n \rightarrow \mathbb{R}$ is the matrix of second-order partials

$$[Hf_{ij}] = \left[\frac{\partial^2 f}{\partial x_i \partial x_j} \right]$$

If $X \subseteq \mathbb{R}^2$, then

$$Hf = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{bmatrix}$$

The Hessian is the transpose of the Jacobian of the gradient:

$$Hf = (J\nabla f)^T.$$

Theorem 6.1.5 (Second partial derivative test)

Let X be an open subset of $\mathbb{R}^n$ and $f : X \rightarrow \mathbb{R}$ whose 2nd-order and lower partials exist and are continuous on X (f is of class C^2). Let $\vec{a} \in X$ be a critical point of f . Then

- If the Hessian $Hf(\vec{a})$ is positive definite, then f has a local minimum at $\vec{a}$.
 - ▶ Easiest way to check is that the leading principal minors (determinant of the top left submatrices) are positive.
 - ▶ This is equivalent to the second derivative $v^T(Hf)v$ being positive.
- If the Hessian $Hf(\vec{a})$ is negative definite, then f has a local maximum at $\vec{a}$.
 - ▶ Easiest way to check is that the 1st leading principal minor is negative, 2nd is positive, 3rd is negative, etc.
 - ▶ This is equivalent to the second derivative $v^T(Hf)v$ being positive.
- If $\det Hf(\vec{a}) \neq 0$ but $Hf(\vec{a})$ is neither positive nor negative definite, then f has a saddle point at $\vec{a}$.

- If the Hessian has both positive and negative eigenvalues, then f has a saddle point at $\vec{a}$.
- Otherwise, the Hessian must be positive semidefinite or negative semidefinite, and the test is inconclusive.

If $X \subseteq \mathbb{R}^2$, let

$$D := f_{xx}(\vec{a})f_{yy}(\vec{a}) - (f_{xy}(\vec{a}))^2 = \det Hf(\vec{a})$$

Then

- If $D > 0$ and $f_{xx}(\vec{a}) > 0$, then f has a local minimum at $\vec{a}$.
- If $D > 0$ and $f_{xx}(\vec{a}) < 0$, then f has a local maximum at $\vec{a}$.
- If $D < 0$, then f has a saddle point at $\vec{a}$.
- If $D = 0$ the test is inconclusive.

(Note that if $f_{xx}(\vec{a}) = 0$ then $D \leq 0$.)

6.2. Taylor polynomials

See <https://youtu.be/V4iscOnH6W8>

The **first-order Taylor polynomial** is just the linear approximation

$$T_1(\vec{x}) = f(\vec{a}) + \nabla f(\vec{a}) \cdot (\vec{x} - \vec{a})$$

Definition 6.2.1 (Second-order Taylor polynomial)

The **second-order Taylor polynomial** for a function $f \in \mathbb{R}^n \rightarrow \mathbb{R}$ at point $\vec{a}$ evaluated at point $\vec{x}$, where $\vec{h} := \vec{x} - \vec{a}$, is:

$$\begin{aligned} T_2(\vec{x}) &= f(\vec{a}) + \sum_{i=1}^n f_{x_i}(\vec{a})h_i + \frac{1}{2} \sum_{i,j=1}^n f_{x_i x_j}(\vec{a})h_i h_j \\ &= f(\vec{a}) + \nabla f(\vec{a}) \cdot \vec{h} + \frac{1}{2} \vec{h}^T (Hf(\vec{a})) \vec{h} \end{aligned}$$

Higher-order Taylor polynomials are not very useful.

6.3. Absolute extrema

Theorem 6.3.1 (Extreme Value Theorem)

Let X be a closed and bounded subset of $\mathbb{R}^n$ and suppose $f : X \rightarrow \mathbb{R}$ is continuous. Then f attains an absolute maximum and an absolute minimum somewhere on X .

Theorem 6.3.2 (Method to find absolute minima and maxima with a constraint)

Let A be a closed bounded subset of $\mathbb{R}^n$, and C be the set of all critical points of f contained within A . Let $S := C \cup \partial A$, the union of C and the boundary of A . Then, the absolute maximum is $\max\{f(\vec{c}) : \vec{c} \in S\}$ and the absolute minimum is $\min\{f(\vec{c}) : \vec{c} \in S\}$.

6.4. Lagrange multiplier

See <https://www.youtube.com/watch?v=8mjcnxGMwFo>

Theorem 6.4.1

If $f(\vec{x}_0) = c$ is an extreme value (absolute max or min) of f on the set $\{\vec{x} : g(\vec{x}) = k\}$ and $\nabla g(\vec{x}_0) \neq 0$, then the following equivalent things are true:

- at $\vec{x}_0$, the level set $\{\vec{x} : f(\vec{x}) = c\}$ is tangent to $g(\vec{x}) = k$.
- $\nabla f(\vec{x}_0) = \lambda \nabla g(\vec{x}_0)$, where $\lambda \in \mathbb{R}$ is called the **Lagrange multiplier**.

Procedure 6.4.2

To find the extreme values of f on the set $\{\vec{x} : g(\vec{x}) = k\}$, solve the system of equations

$$\begin{aligned}\nabla f(\vec{x}) &= \lambda \nabla g(\vec{x}) \\ g(\vec{x}) &= k.\end{aligned}$$